Max Ham

Lead Machine Learning Engineer
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Summary

Machine learning platform engineer experienced in building and operating production AI/ML systems across enterprise environments. Designed event-driven services supporting 3M+ transactions per day, deployed 100+ models on Kubernetes, and established reusable inference, CI/CD, and observability capabilities. Combines hands-on work in Azure, Databricks, GPU model serving, and distributed data platforms with cross-functional technical leadership.

Work Experience

Lead Machine Learning Engineer 2026 - Present
Layer 6 AI Toronto, ON

- Architected an enterprise LLM gateway for OpenAI, Anthropic, and self-hosted models, centralizing authentication, authorization, rate limiting, token-usage tracking, and scalable API infrastructure.
- Established production model-serving infrastructure with NVIDIA Triton Inference Server, GPU-optimized containers, and reusable deployment patterns for enterprise AI/ML workloads.
- Designed a reusable MCP-based agentic AI platform with patterns for model and tool integration, orchestration, RAG, context engineering, reusable skills, and human-in-the-loop workflows.
- Set AI platform observability standards with Datadog and Dynatrace, implementing centralized dashboards, logging, and telemetry across AI workloads.
- Led 15+ engineers across network, architecture, and platform teams to deliver active-active disaster recovery for a Tier-1 predictive platform supporting $100M+ in production use cases; mentored engineers and contributed to hiring.
- Translated enterprise AI strategy into knowledge discovery, codebase intelligence, workflow automation, data analysis, IT operations, and document intelligence use cases, improving organizational workflow efficiency by 35%.

GenAI Platform Technology Lead 2024 - 2026
Canadian Imperial Bank of Commerce Toronto, ON

- Designed and built a scalable, event-driven platform for synchronous and asynchronous AI/ML workloads, supporting 3M+ transactions per day and large-scale document processing.
- Deployed and operated 100+ GenAI/ML models on Kubernetes across 300+ document types; enabled Databricks Apps human-in-the-loop workflows that contributed to over $15M in customer profit realization.
- Led a cross-functional team of 20+ engineers across two platforms, establishing CI/CD and DevSecOps practices and automating secure ServiceNow deployments.
- Built GPU-enabled container environments for ML workloads, configuring CUDA dependencies and optimized runtime images for model development and deployment at enterprise scale.
- Architected secure Azure workload and client identity using OAuth 2.0, OIDC, federated credentials, managed identities, certificate authentication, mTLS, and token exchange across Kubernetes and enterprise identity providers.
- Launched an MLflow-based CI/CD pipeline using Databricks and Azure CLIs in GitHub Actions to deploy machine learning models, reducing team workload by 100%.

Senior Application Developer 2024 - 2024
Canadian Imperial Bank of Commerce Toronto, ON

- Implemented reusable enterprise data pipelines handling petabytes across concurrent migrations, generating $600K+ in revenue within three months.

Relevant Skills

ML / AI: MCP, RAG, Agentic AI, NVIDIA Triton Inference Server, CUDA, OpenAI, Azure Machine Learning, Generative AI, ML Model Serving, Databricks, AI Foundry, AI Search, Document Intelligence
Platform: Kubernetes, Helm, Docker, Linux OS, API Management, Container Registry, Service Bus, Solace, OAuth 2.0, OIDC, Event Grid/Hub, F5, Akamai, Unity Catalog, PingFederate, App Service Plan, RBAC, NSG, Virtual Network
DevOps: GitHub Actions, Terraform, Jenkins, HashiCorp Vault, ServiceNow, Maven
Technical: bash, Python, Java, SpringBoot, DQL, .NET, KQL, SQL, JQL, Spark, Scala, TypeScript, JavaScript